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Project Title:	Non-Intrusive Measurement of Thermophysical Properties of Liquids by Electrostatic-Acoustic Hybrid Levitation		
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Task Description:	The main objective is to devise non-contact methods of carrying out measurements of thermophysical properties of highly viscous liquids. The eventual aim is to be able to measure the properties of liquids in undercooled states for which non-contact techniques are essential for minimizing the disturbance to the metastable thermodynamic state. The objective of this research is to devise techniques to determine the thermophysical properties of liquids in their undercooled states. These properties will be inferred from measurements of the characteristics of thermocapillary flow induced by spot heating a levitated sample. The undercooled state of a liquid is achieved under controlled quiescent conditions whereby a liquid can be cooled below the freezing point while still maintaining the liquid state. Such states are found naturally or can be created by levitating a liquid drop without a container. Liquids in undercooled states also exist in some industrial applications such as liquid-metal spray deposition. For the purpose of metal coating, this process is considered environmentally friendlier than the traditional electroplating that produces hazardous waste. For various such processes where undercooled states are found, it is important to know the properties of the materials. The thermophysical properties include thermal conductivity and viscosity that are fundamentally important in dealing with the liquid forms of many materials in industrial applications. The ongoing research will focus on the development of techniques that will allow the non-intrusive measurement of the properties that have been out of scientific reach due to the undercooled state being destroyed by contact with container walls or instrumentation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The basic thermodynamics of the undercooled state is little understood. The measurement techniques developed will be useful in better understanding of various unknowns in this arena.
	Spheroidal Approximation for Finite-Amplitude Highly-Viscous Axisymmetric Drop/Bubble Free Shape Relaxation The problem of a highly-viscous axisymmetric shape relaxation towards a sphere for a drop freely suspended in another immiscible fuid has been considered, and the relaxation rate has been calculated. Our earlier work (Ohsaka, Rednikov, Sadhal, S, & Trinh, 2002) on the noncontact measurement of viscosity of highly-viscous liquids by shape relaxation has led to the need for a refined model, and consequently, the current development has been conducted. The only heuristic element of the analysis has been the premise that the drop retains a spheroidal shape (oblate or prolate) during an entire process of relaxation (spheroidal approximation/ approach), while an exact problem was solved in all other respects. Simple expression of the solution is not the only use for such an approach. Another one is that a certain universality of the result presentation is achieved in the sense that the relaxation rate is expressed in terms of only one shape parameter, although at the expense of making the results not exact. The viscosity ratio has been varied covering all possible cases (from a bubble to a drop in the air). Mathematically, the problem has been treated in such a way that the heuristic element enters through the surface normal stress balance, which is satisfied only approximately, whereas all other boundary conditions are satisfied exactly. The relaxation rate is obtained by approximating the normal stress balance in a reasonably good manner. As the analysis is only approximate, this procedure is characterized by some degree of arbitrariness. Following Taylor (1964), two distinct procedures have been adopted, which are referred to as Approximations I and II. The results of these two approximations are generally different, as the spheroidal approach is not exact. A relative difference between them has been used as a quantitative measure of the inconsistency of the spheroidal approximation. Both the relaxation rate and the inconsistency are functions of the shape parameter (such as the aspect ratio) and the viscosity ratio only. All in all, we conclude that the problem of drop shape relaxation in the highly viscous regime is quite amenable to the spheroidal approximation, with the inconsistency values generally being reasonably small. The inconsistency generally grows with deformation (oblate or prolate), and is higher for large inside-to-outside viscosity ratios than for small ones. In other words, the spheroidal approximation tends to work better when the fluid with lower dynamic viscosity is on the inside, than outside. In particular, a bubble is found to be appreciably more amenable to the spheroidal approach than a drop in air. The latter is the case where the spheroidal approximation appears to work the worst (inconsistency values are maximum). For example, the inconsistency is of the order of 0.5% for an oblate bubble with the aspect ratio a/b = 2, but about 7% for a drop of the same geometry in air. For a prolate bubble with a/b = 1/2, the inconsistency is about 2%, but about 8% for a similar drop in air. Here a and b are the dimensionless equatorial and polar radii of the spheroidal drop, such that a2b = 1 (volume conservation condition). An interesting phenomenon exists due to the inconsistency value going down to zero at a certain value of the inside-to-outside viscosity ratio (always with the less viscous fluidinside), which depends on the aspect ratio and is found to decrease as the deformation (either oblate or prolate) is increased. Such a decrease is more pronounced in the oblate case. As a consequence, we found that the spheroidal approach works especially wellfor the relaxation of appreciably deformed oblate bubbles. The inconsistency being zero merely means that the results of Approximations I and II happen to coincide. It does not mean that the spheroidal approach becomes exact. Nonetheless, it is generally true that the spheroidal approach works especially well in a vicinity of such a point in the parameter space (the parameters being the viscosity ratio and the aspect ratio). Perhaps, this is why it turns out to work better when the fluid inside the drop is less viscous than the one outside: all points of zero inconsistency correspond to small values of the inside-to-outside viscosity ratio. A more detailed inspection of the resulting normal stress mismatch and its comparison with the minimal Laplace pressure value in the system reveals that the inconsistency values as low as 10% should in fact already be regarded large, for the distortion of the spheroidal shape might already be significant. If 10% is adopted as the upper bound on the admissible inconsistency values, then for a drop in air and other cases of high inside- to-outside viscosity ratio, the spheroidal consideration is limited to moderate deformations of the order of a/b = 2 and a/b = 1/2. For a/b = 3 and a/b = 1/3, non-spheroidal tendencies may be too signficant. Again, results are better when a less viscous fluid is on the inside. For example, in the case of a bubble, no reason has been found why the spheroidal approach cannot be applied to deformations as high as a/b = 4 or a/b =1/4, or maybe even higher (although possible instabilities, if any, leading to a non-spheroidal shape evolution cannot be detected within the spheroidal approach). What becomes of the spheroidal approximation in the small-amplitude limit, when the drop shape is close to a sphere has

also been examined. As the leading-order ('linear') approximation is trivial within the present setup, of interest was mainly the next-order ('quadratic') approximation for the relaxation rate. While within the linear approximation (for the quadrupole mode, n = 2), the drop is trivially a spheroid, this is no longer the case within the quadratic order, where the mode n = 4 is also present whose amplitude is not expected to be in the necessary correlation with the n = 2 amplitude for the shape to evolve exactly as a spheroid. However, it has been found that enforcing a spheroidal shape results in the scaled (with the small amplitude [(a2 – b2) << 1] inconsistency not exceeding 2% for all viscosity ratios. Furthermore, for the inside-to-outside viscosity ratio approximately equal to 0.432, the inconsistency becomes equal to zero, which signals that the spheroidal approach is exact at this point within the quadratic approximation (unlike the case of ¯nite deformation, where it is not exact even at such a point). A straightforward perturbation analysis up to and including the quadratic order would be useful in order to better understand these findings.

Task Progress:

An important consequence deduced by Mo, Zhou & Yu (2000) from an overall heuristic spheroidal model for drop shape relaxation by Maffettone & Minale (1998) is that, in our present terms, the quantity (a2 – b2) distinguishes itself in that it evolves formally according to the linear approximation even for significant drop deformations (here referred to as the MMMZY result). Incidentally, this finding inspired the form the relaxation rate results have been presented in the present paper (in terms of Relative Rate Excess). The approach used here is heuristic to a lesser extent, and thus we have been in a position to check up how well the MMMZY prediction works. It proves to work quite satisfactorily in fact. For example, in the oblate case with a/b = 2, the discrepancy with the present results is only about 12% for a bubble. Yet, this is well above the inconsistency value estimated at about 0.5%, and thus the present paper reliably yields a more accurate result for the relaxation rate in this case. On the other hand, for a drop in air, the discrepancy falls below the calculated inconsistency value of about 7%. In such a case, no improvement over the MMMZY prediction is achieved here, although the inconsistency estimate is still a valuable additional finding. In the prolate case with a/b = 1/2, we have a 19% discrepancy with the MMMZY prediction for a bubble, which should now be compared with a 2% inconsistency. For a drop in air, the numbers are 20% and 8%, respectively. Interestingly enough, except for sufficiently high values of the inside-to-outside viscosity ratio, the relaxation rate in terms of (a2– b2) tends to be higher than the corresponding linear relaxation rate in the oblate case, and lower in the prolate case. For high viscosity ratios, and for a drop in air in particular, the actual relaxation rate in terms of (a2 – b2) is higher than the linear one both in the oblate and prolate cases for large enough deformations.

Non-Contact Thermal Diffusivity Estimation by Thermocapillary Flow Analysis and Measurement in a Levitated Horizontal Thin Disk

Experiments with thermocapillary flow and temperature measurements in a horizontally levitated, laser-heated thin glycerin disk have been successfully carried out. Such a heated disk has the advantage of a relatively low gravitational potential over the thickness, thus mitigating the buoyancy effects, and helping isolate the thermocapillary driven flows. For the purpose of predicting the thermal properties from these measurements, it was necessary to develop a theoretical model of the thermal processes. Such a model has been being developed, and based on the observed shape, the thickness is taken to be a minimum at the center with a gentle parabolic profile at both the top and the bottom surfaces. This minimum thickness is much smaller than the radius of disk drop and the ratio of thickness to radius becomes much less than unity. It is heated by laser beam in normal direction to the edge. A general three-dimensional momentum equation is transformed into a two-dimensional vorticity equation. For the very viscous small liquid drop, the Stokes equation can be used to describe the fluid motion with viscous force and pressure force. Considering boundary conditions on the surface where the surface stresses are balanced with the thermocapillary force corresponding to temperature gradient, we take an average over thickness of disk drop and obtain a two-dimensional momentum equation which includes surface tension terms. To avoid difficulties with handling pressure terms, it is formulated in terms of the vorticity and some additional terms due to a dimpled surface. This vorticity equation has temperature gradient (corresponding to a Marangoni number) in the plane of the disk which brings about its coupling with the energy equation. In the same way, the three-dimensional energy equation is averaged over the disk thickness. With convection boundary condition at the surfaces, we integrate a general three-dimensional energy equation to get an averaged two-dimensional energy equation which has convection terms, conduction terms, and additional source terms corresponding to a Biot number.

The Biot number evaluation entails the calculation of exterior convective heat transfer, including the flow that is predominantly acoustically driven. This is different from the usual convective heat transfer, in that the acoustic streaming is effectively the equivalent of a slip velocity at the liquid-gas interface. The streaming flow field is estimated by means of approximating the flattened drop as a spheroid for which the analysis has been conducted under the current project (see Rednikov & Sadhal, 2004). The disk diameter is taken to be the same as the diameter of the oblate spheroid. However, disk thickness at along the vertical axis is not taken to be the oblate spheroid thickness. Instead, the disk curvature at the edge is equalized with the oblate spheroid while the latter's thickness at the axis is allowed to be freely determined by these two parameters (diameter and curvature). Although the oblate spheroid turns out to be thicker than the dimpled disk, it is a better approximation from a streaming standpoint. The convective transport equations for the exterior flow are solved semi-numerically using Legendre series expansions corresponding to the analytical solution for streaming about the oblate spheroid. For far-field and surface temperatures in the range corresponding to the experiments, the Nusselt number (and hence the Biot number) is found as a function of position on the surface.

For the internal flow as well, a numerical approach is used to solve these steady state governing equations in cylindrical coordinates. These governing equations are discretized into finite difference equations by using central difference approximation. In a cylindrical coordinate system, there could be many singularities at the center. The calculations yield the temperature distribution and the thermally-driven flow field. These results have been used to formulate a model that, in conjunction with experiments, enables the development of a method for the non-contact thermophysical property measurement of liquids. In particular, comparisons of the experimentally observed and the predicted flow fields, and adjustment of the unknown flow parameters to match the experiment with the theory yields an estimate of the thermal diffusivity.

Distinguished Keynote Lecture "Measurement of Thermophysical Properties of Highly Viscous Liquids by Non-Contact Methods Using Acoustic and Electrostatic Levitation," M.V. Krishna Murthy Distinguished Lecture, Sixth ISHMT-ASME Heat and Mass Transfer Conference, January 5-6, 2004, Kalpakkam India..

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